

$$\textcircled{1} \quad f(x) = \frac{x-1}{\sqrt{2x-x^2}} \quad x \in X = \left[\frac{1}{2}, \frac{3}{2}\right] \quad \text{Og } \underline{37}$$

$$(i) \quad \phi \text{ für Newton} \quad \left(\phi(x) = x^3 - 3x^2 + 3x \right)$$

$$(ii) \quad \underline{\phi'(x) \geq 0 \quad \forall x \in X}$$

$$(iii) \quad \phi \text{ Kontraktion auf } X \\ \text{mit } \alpha \leq \frac{3}{4}$$

$$\Gamma_{17} \quad \underline{\phi(X) \subset X} \quad \phi \text{ monoton w.}$$

$$\phi\left(\frac{1}{2}\right) \in \left[\frac{1}{2}, \frac{3}{2}\right]$$

$$\phi\left(\frac{3}{2}\right) \in \left[\frac{1}{2}, \frac{3}{2}\right]$$

$$[(iv) \quad |\phi'(\xi)| \leq \frac{3}{4} \quad \forall \xi \in \left[\frac{1}{2}, \frac{3}{2}\right]$$

$$(i) \quad f(x) = \frac{x-1}{\sqrt{2x-x^2}}$$

$$\phi(x) = x - \frac{f(x)}{f'(x)}$$

$$\begin{aligned} f'(x) &= \frac{1 \cdot \sqrt{} - (x-1) \frac{2-2x}{2\sqrt{}}}{\sqrt{}^2} \\ &= \frac{\sqrt{1^2} - (x-1)(1-x)}{\sqrt{1^3}} \\ &= \frac{\cancel{x}x - \cancel{x}^2 - (\cancel{x} - \cancel{x}^2 - 1 + \cancel{x})}{\sqrt{1^3}} \\ &= \frac{1}{\sqrt{1^3}} \end{aligned}$$

$$\begin{aligned} \phi(x) &= x - \frac{\frac{x-1}{\sqrt{}}}{\frac{1}{\sqrt{1^3}}} \\ &= x - \frac{x-1}{\cancel{\sqrt{}}} \sqrt{1^3}^2 \\ &= x - (x-1)(2x-x^2) \end{aligned}$$

$$\phi(x) = \dots = \underline{x^3 - 3x^2 + 3x}$$

$$(ii) \quad \underline{\phi'(x) \geq 0 ; \quad x \in \left[\frac{1}{2}, \frac{3}{2}\right]}$$

$$\begin{aligned} \phi'(x) &= 3x^2 - 6x + 3 \\ &= 3(x^2 - 2x + 1) = 3(x-1)^2 \geq 0 \end{aligned}$$

(iii) ϕ Kontraktion auf $X = [\frac{1}{2}, \frac{3}{2}]$:

$$(a) \quad \phi(X) \subset X$$

$$(b) \quad |\phi(x) - \phi(y)| \leq \frac{3}{4} |x - y|$$

$$\forall x, y \in X$$

(a) aus (iii) $\phi' \geq 0$ auf X

$\Rightarrow \phi$ monoton wachsend auf X

$$\phi\left(\frac{1}{2}\right) = \left(\frac{1}{2}\right)^3 - 3\left(\frac{1}{2}\right)^2 + 3\frac{1}{2}$$

$$= \dots = \frac{7}{8} \in \left[\frac{1}{2}, \frac{3}{2}\right]$$

$$\phi\left(\frac{3}{2}\right) = \dots = \frac{9}{8} \in \left[\frac{1}{2}, \frac{3}{2}\right]$$

b) MWS :

$$|\phi(x) - \phi(y)| \leq |\phi'(\xi)| |x - y|$$

$$\max_{\xi \in X} |\phi'(\xi)| \leq \frac{3}{4}$$

$$\phi'(\xi) = 3(\xi - 1)^2$$

$$\phi'\left(\frac{1}{2}\right) = \frac{3}{4}, \quad \phi'\left(\frac{3}{2}\right) = \frac{3}{4}$$

$$\phi''(z) = 6(z-1) = 0$$

$$\Rightarrow z = 1$$

$$\phi'(1) = 0$$

$$\begin{aligned} \max_{z \in [\frac{1}{2}, \frac{3}{2}]} |\phi'(z)| &= \max \left(\overset{\frac{3}{4}}{\underbrace{|\phi'(\frac{1}{2})|}}, \overset{\frac{3}{4}}{\underbrace{|\phi'(\frac{3}{2})|}}, \overset{=0}{\underbrace{|\phi'(1)|}} \right) \\ &= \frac{3}{4} \end{aligned}$$